

Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

In lab 2 I learned the basics of how to set up a verilog project, how to connect an input and output using verilog, and how the class generally runs. We also talked about the implementation of boolean functions (like AND and XOR) in verilog, though we didn't actually put this into practice.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

1. Set up project files
 - a. Download the relevant lab files
 - b. Create a new project in Vivado
 - c. Go through the setup dialogues, making sure to select the correct lab files & the correct board type
2. Write code & run simulations until the simulations show the desired results
 - a. Write & edit code
 - b. Select Run Simulation > Run Behavioural Simulation
 - c. Check the waveform of the simulation to see if you got the desired results
 - d. If no, repeat this step. If yes, continue to step 3.
3. Connect the board & run the project
 - a. Select Generate Bitstream and hit Yes on the dialog to agree to the implementation process. Then select OK on the second dialog to confirm settings. This is kind of like compiling code.

- b. Once the previous step is done, select Open Hardware Manager, plug in the board, and select Open Target, Auto Connect, Program Device, and Program. This basically sends the “compiled” code to the board and runs it.
- c. Check if the board acts the way you want it to

2 - What is the value in looking at the elaborated design schematic?

This helps us visualize what's going on inside the board and allows us to have another model to check the code against.

3 - Why should we simulate our designs frequently? What does the simulation do?

Simulation allows us to test our designs and make sure everything is working properly before we put it on the hardware. Doing this frequently helps identify which part of the code might be causing a problem, instead of trying to track down a bug from code you wrote hours ago and didn't test.